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Metage2Metabo PostAViz: a Python package for exploring, visualising, and comparing the metabolic potential of microbial communities

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Summary

Microbial communities comprise up to several thousand distinct microbial populations, each defined by its genomic DNA sequences and sharing a common habitat and environmental conditions. The term *microbiome* describes the holistic concept encompassing these microorganisms and their environment (Berg et al., 2020). Characterising the populations within samples and understanding their roles and interactions in the microbiome requires a combination of experimentation, high-throughput data acquisition and computational modelling (Klitgord & Segrè, 2011).

The role of microbial populations can be abstracted through the study of their *metabolism*, encompassing the full set of biochemical reactions performed by cells. Within a microbiome, metabolic functions often display both redundancy and complementarity: distinct microbial compositions may support similar overall functions. This functional overlap motivates the systematic inference of metabolic potential. Because the genome encodes the information underlying metabolism, it can be used to estimate the metabolic capabilities of microorganisms, typically represented as *metabolic networks*. These networks form the basis of *metabolic models*, which integrate additional data within mathematical or computational frameworks to predict the behaviour of microbial communities under specific environmental conditions.

Advances in experimental strategies and large-scale DNA sequencing in metagenomics now enable the analysis of increasing numbers of samples. Scalable approaches are therefore required to explore potential behaviours and interactions among microbial populations across environments. Comparing predictions across multiple samples is essential for generating actionable hypotheses, but doing so requires integration of sample metadata, statistical analysis, and careful interpretation of modelling outputs. Effective visualisation and customisable exploration tools are therefore important to make metabolic modelling results accessible and usable for end users.

Statement of need

`Metage2Metabo-PostAViz` (M2M-PostAViz) is a Python package that performs analyses on the predictions generated by the metabolic-modelling tool `Metage2Metabo` (M2M) (Belcour et al., 2020). Concretely, it takes as input multiple metabolic network files (one per microbial population of the modelled community) together with a set of seed metabolites that describe the environmental conditions. M2M encodes these networks as a discrete model in the symbolic AI framework of Answer Set Programming (ASP) and computes which metabolites are reachable from the provided seeds. The inference accounts for the complementarity between networks, allowing metabolites produced by one population to be used by others, thereby modelling potential metabolic interactions within the community.

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When working with cohorts of hundreds or thousands of samples, M2M must be executed separately for each sample. The resulting outputs then need to be collected and analysed across samples to enable comparison and interpretation of model predictions. For each community, M2M produces several data frames describing the role of each microorganism with respect to the community-level functions. Proper cross-sample comparison therefore requires aggregating these outputs and integrating sample metadata, such as microbial composition, lifestyle variables, or clinical information.

`M2M-PostAViz` addresses this need by integrating M2M outputs across samples together with associated metadata into a unified data structure that supports systematic comparison. Results and metadata are stored in the parquet columnar format, enabling efficient disk-based storage and fast data access while limiting memory usage. This design allows the exploration of hundreds to thousands of samples without degrading user experience. In addition, M2M-PostAViz provides a preprocessing step in which outputs are structured and stored in advance, so that subsequent exploratory sessions can be launched efficiently.

The package offers a visualisation interface for custom plot generation and statistical analyses, enabling stratified exploration of metabolic capabilities across samples or groups. Exploration can be conducted at multiple levels: individual metabolites that may or may not be producible across samples; microorganisms, whose predicted behaviour may vary depending on interactions with other community members; and broader summaries of community-level functions. The tool operates as a local web-based application.

Overall, `M2M-PostAViz` was designed to address three complementary needs: (i) enabling the comparison of metabolic modelling results across large metagenomic cohorts by structuring and aggregating per-sample M2M outputs; (ii) integrating modelling predictions with additional data sources, such as microbial community composition and sample metadata, to support statistically informed analyses; and (iii) making these analyses accessible to non-specialists through a graphical user interface that supports customisable workflows.

State of the field

Metabolic modelling is widely used to investigate the roles of microorganisms in microbial communities, including which molecules they can produce and which interactions may occur under particular environmental conditions (Cerk et al., 2024). Many models rely on integer linear programming optimisations (García-Jiménez et al., 2021) and some alternatives use Boolean abstractions (Belcour et al., 2020; Frioux et al., 2018) or probabilistic approaches (Bernstein et al., 2019) to provide predictions on community behaviours. Key questions are scalability to large communities, and also integration of these predictions when analysing many samples or community compositions, which is more and more frequent as large metagenomic cohorts get published (Asnicar et al., 2025).

In practice, each microorganism is abstracted by a collection of biochemical reactions it may perform according to its genomic information, thus forming a network connecting transformed molecules (Cerk et al., 2024). A community of microorganisms is therefore represented as a collection of such networks, referred to as genome-scale metabolic networks. A metabolic model will provide predictions on the possible behaviour of microorganisms and communities in defined simulation conditions. Boolean abstractions, as used by M2M, ensure the scalability of predictions; although the model is qualitative and does not quantify interactions, such methods are considered a reasonable proxy for more quantitative models (Kruse & Ebenhöh, 2008) (Figure Figure 1A).

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Software design

Data integration and storage

`M2M-PostaViz` is implemented in Python (3.12). Each sample must first be processed with `M2M`. The inputs to `M2M-PostAViz` are the directories generated by `M2M` for each sample, which contain multiple tabulated and JSON files. Additional inputs include microbial relative abundances (used to weight predictions), sample metadata (e.g. lifestyle or clinical variables), and optionally metadata describing metabolites and microbial taxonomy. These inputs are provided as tabular files.

Integrating these heterogeneous data sources across hundreds or thousands of samples can be computationally demanding, particularly since each sample corresponds to a separate output directory. Moreover, users typically explore the same dataset across multiple sessions. To avoid repeated parsing and integration of raw outputs, M2M-PostAViz implements a preprocessing step that consolidates and structures all required data into a unified representation.

The processed data are stored using the parquet columnar format. This enables efficient disk-based storage and fast, selective data access while limiting memory usage. As a result, the initial preprocessing step may be time-consuming, but subsequent application runs directly load the structured data, substantially reducing startup time and improving responsiveness during interactive exploration.

Data manipulation and storage rely primarily on pandas (McKinney, 2010; team, 2020), polars (The Polars development team, 2024), and pyarrow (Richardson et al., 2026) for tabular processing and parquet integration.

Application content

The application opens as a multi-tab browser page. The first tab provides an overview of the dataset and enables initial analyses that can be customised using metadata variables. Two additional tabs provide analyses centred on microbial roles and metabolites, respectively (see below), and a final tab summarises the metadata and allows users to customise variable types to fine-tune analyses and plots. Statistical analyses are performed using scipy (Virtanen et al., 2020) and statsmodels (Seabold & Perktold, 2010). Visualisations are generated with plotly (Inc., 2015), matplotlib (Hunter, 2007), and seaborn (Waskom, 2021), and the web interface is implemented using shiny for Python (The Shiny development team, 2024) (Figure Figure 1B).

Exploration of microorganism roles

The second tab of the application focuses on the role of microorganisms in the production of metabolites across samples. The same species and thus the same metabolic network can appear in several samples, but behave differently because of possible interactions with other microbial populations. In addition, microbial species can be grouped according to their taxonomy, enabling users to consider not only the role of a metabolic network but the role of all those falling in taxonomic groups of different levels (*e.g.* phylum, family, genus). Again, sample metadata variables can refine the analysis, enabling users to compare groups or filter certain samples.

Exploration of metabolite production across samples

The third tab of the application is dedicated to the analysis of metabolites, which can also be grouped into families if a suitable ontology is provided (Aite et al., 2018; Caspi et al., 2019). Focusing on metabolites enables a targeted analysis in order to compare samples or groups of samples on specific metabolic functions.

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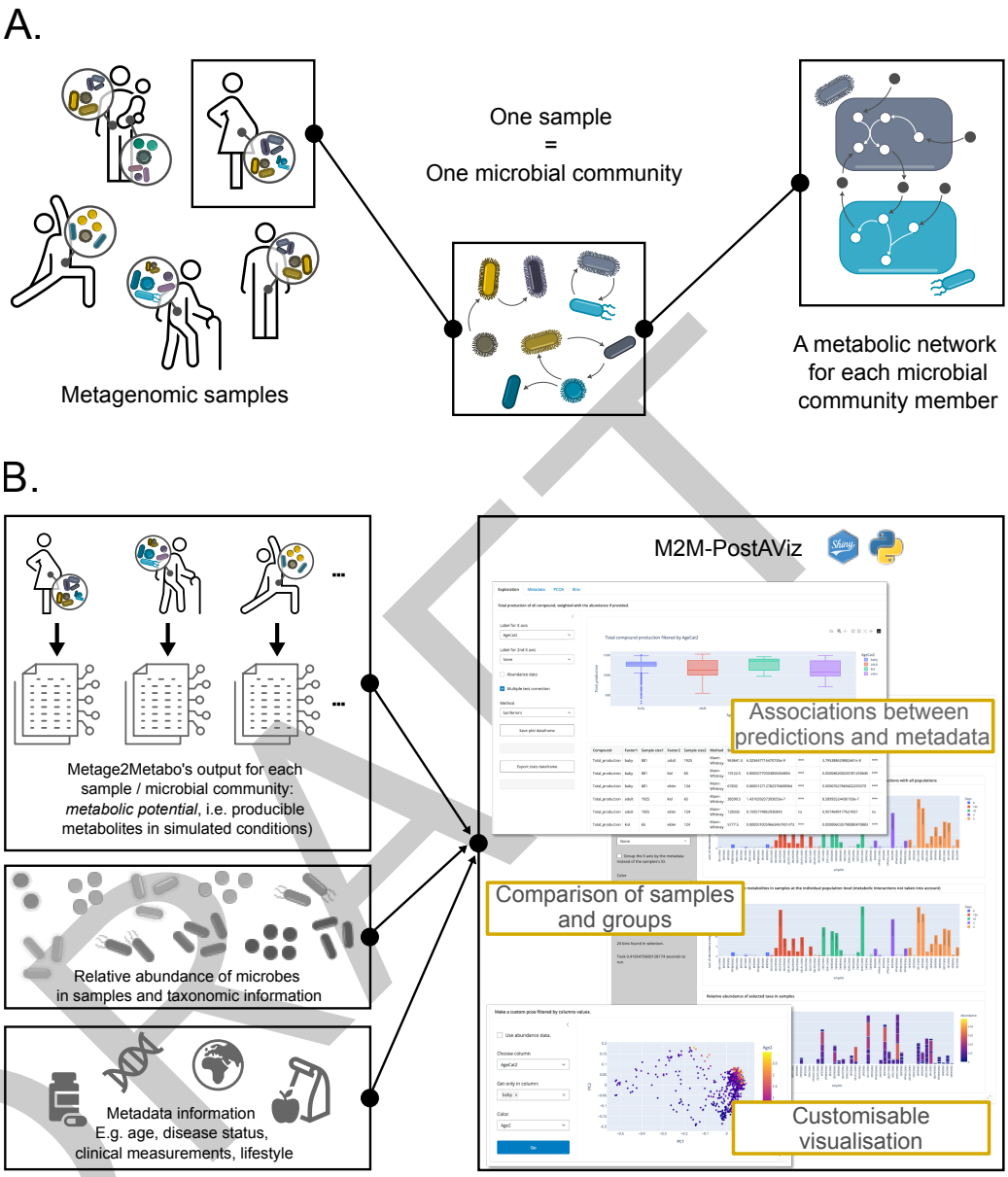


Figure 1: *Illustration of the context and functionalities of Metage2Metabo-PostAViz.* **A.** Metagenomic context. Samples consist of microbial communities, each composed of several interacting microbial populations. These populations are represented as metabolic networks used to predict community behaviour and interactions within dedicated modelling frameworks. **B.** Functionalities of M2M-PostAViz. Inputs include Metage2Metabo outputs for each sample, metadata, relative abundances, and taxonomic information associated with the samples. The application enables comparison of the metabolic potential of samples and exploration of their associations with metadata through a graphical user interface.

Research impact statement

`M2M-PostAViz` builds on the widely used M2M framework, which has been cited more than one hundred times and remains actively maintained, by addressing key barriers that currently limit the broader adoption of large-scale metabolic modelling. While M2M enables powerful predictions of metabolic complementarity and community-level functions, its complexity and the volume of generated outputs require substantial expertise, restricting its use to a limited and expert audience.

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M2M-PostAViz overcomes these limitations by providing an interactive and user-friendly graphical interface that enables systematic exploration, comparison, and interpretation of M2M outputs across large metagenomic datasets consisting of up to thousands of samples. The tool integrates metabolic modelling results with microbial community composition and associated metadata, facilitating multi-dimensional analyses at scales that were previously difficult to achieve.

By lowering technical barriers, M2M-PostAViz broadens access to metabolic modelling approaches and enables their application by non-specialists. To promote transparency, reproducibility, and adoption, M2M-PostAViz is accompanied by an extensive tutorial and curated test datasets, supporting users in applying metabolic modelling analyses to diverse microbial ecosystems.

AI usage disclosure

AI tools (GitHub Copilot and GPT-5) were used occasionally for code generation, particularly for debugging, and for grammar and language checks of the manuscript. All suggestions were reviewed and validated by the human authors.

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